

VIKTOR CHALLENGE · TUM.AI · BUILD THE ROUTER

Bayern CodeWerk

**–37% of the bill at 99.3% of current outcome
quality — with error bars, and we show you
where it breaks**

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OBJECTIVE

The trade-off we chose

Input cost per task, holding outcome quality. It is the only axis this export supports honestly: no timings ship, so latency is unmeasurable, and no quality labels ship, so quality has to be constructed — we treat it as a *constraint*, not a target.

THE SIGNAL

What the router looks at

Structure in the traces

Risk signals, not prompt wording. The router reads the tool signature, whether a stakes tool was called, whether the last result was an error, and whether it is stuck in a retry loop. It starts on a price ladder and moves up or down by named steps. **Every decision reports which rules fired.**

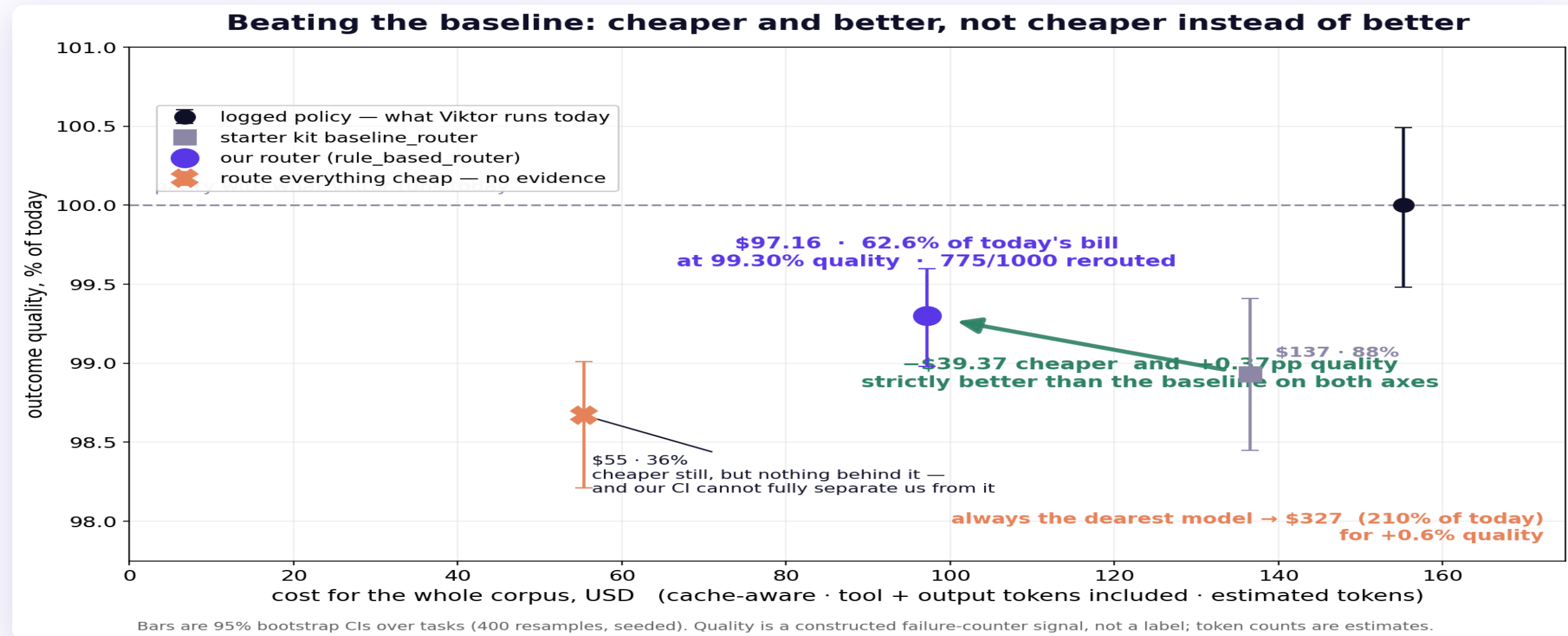
Shown on a real trace

Viktor does not route per task today — a human picks one model per workspace. So recurring cron jobs drifted across price tiers on their own: **90 job families, 69 of them on two or more tiers.**

Same job, different model. That accident is what lets us measure a model that never ran.

THE ONE CHART

Cost–quality frontier (full public dataset · n=1000 · cache-aware)



vs the starter baseline: \$39 cheaper AND +0.4pp quality — strictly better on both axes, not cheaper instead of better.

logged \$155.23 · baseline \$136.53 (88%) · **ours \$97.16 (62.6%)** at 99.30%, 95% CI [98.98, 99.60] · 775/1000 rerouted, all nine models still in play · always-dearest \$326.65 (210%) for +0.6%

The off-policy estimate — and where it breaks

Method — matched same-job comparison. Recurring crons drifted across price tiers on their own, so difficulty is held fixed *by construction*, not by a similarity score. Quality is a failure-counter signal over 10,422 tool calls; 95% bootstrap CIs, 400 resamples, seeded.

Cache trap — priced cache-aware throughout; on this export each row is one complete task, so there is no shared prefix to lose. We also count what the kit omits — tool tokens and output — so **the true bill is \$155, not \$87**, and every percentage here is of that larger number.

Weakest point — a fluent, confident, *wrong* answer with no failed tool call scores **perfectly**. That bounds every quality claim here. And our CI overlaps “route everything cheap” ([98.98, 99.60] vs [98.21, 99.01]) — we cannot claim to beat the loophole on *measured* quality. What separates us is keeping all nine models in play and escalating on risk: a design argument, not a measured one.

CLOSE

Viktor has no per-task routing today. Adding it is close to free.

**\$155.23 → \$97.16 · 62.6% of the bill at
99.3% of current quality**

beats the starter baseline on cost *and* quality · 775/1000 tasks rerouted

Known weakness: our estimator cannot separate us from routing everything cheap on quality alone — we say so before you ask. **Next:** judge-model rescoring on the segments where the two policies disagree, to turn a design argument into a measured one.